

4.9 Solution: (a) We can directly write the scalar potential inside and outside of the dielectric sphere as

$$\Phi_{\text{out}} = \sum_{l=0}^{\infty} \left(\frac{q}{4\pi\epsilon_0} \frac{r_{<}^l}{r_{>}^{l+1}} + \frac{A_l}{r^{l+1}} \right) P_l(\cos \theta), \quad \Phi_{\text{in}} = \sum_{l=0}^{\infty} B_l r^l P_l(\cos \theta).$$

By the boundary conditions

$$-\epsilon \frac{\partial \Phi_{\text{in}}}{\partial r} \Big|_{r=a} = -\epsilon_0 \frac{\partial \Phi_{\text{out}}}{\partial r} \Big|_{r=a}, \quad -\frac{1}{a} \frac{\partial \Phi_{\text{in}}}{\partial \theta} \Big|_{r=a} = -\frac{1}{a} \frac{\partial \Phi_{\text{out}}}{\partial \theta} \Big|_{r=a},$$

we have the following equations for the coefficients of the scalar potential expansion, for all l ,

$$\epsilon_0 \left(\frac{q}{4\pi\epsilon_0} \frac{la^{l-1}}{d^{l+1}} - \frac{(l+1)A_l}{a^{l+2}} \right) = \epsilon l B_l a^{l-1}, \quad \frac{q}{4\pi\epsilon_0} \frac{a^l}{d^{l+1}} + \frac{A_l}{a^{l+1}} = B_l a^l,$$

which can be solved as

$$A_l = -\frac{(\epsilon - \epsilon_0)l}{l\epsilon + (l+1)\epsilon_0} \frac{a^{2l+1}}{d^{l+1}} \frac{q}{4\pi\epsilon_0}, \quad B_l = \frac{(2l+1)\epsilon_0}{l\epsilon + (l+1)\epsilon_0} \frac{1}{d^{l+1}} \frac{q}{4\pi\epsilon_0}.$$

(b) Given Φ_{in} , we have

$$\mathbf{E}_{\text{in}} = -\nabla \Phi_{\text{in}} = -\frac{\partial \Phi_{\text{in}}}{\partial r} \hat{r} - \frac{1}{a} \frac{\partial \Phi_{\text{in}}}{\partial \theta} \hat{\theta}.$$

Nere the center of the sphere, $r \approx 0$, we only need to consider the $l = 1$ term in the potential, and

$$\mathbf{E} = -\frac{q}{4\pi\epsilon_0 d^2} \frac{3\epsilon_0}{\epsilon + 2\epsilon_0} (\cos \theta \hat{r} - \sin \theta \hat{\theta}) = -\frac{q}{4\pi\epsilon_0 d^2} \frac{3\epsilon_0}{\epsilon + 2\epsilon_0} \hat{z}.$$

(c) In the limit of $\epsilon/\epsilon_0 \rightarrow \infty$, we can see that

$$A_l \rightarrow -\frac{a^{2l+1}}{d^{l+1}} \frac{q}{4\pi\epsilon_0}, \quad B_l \rightarrow 0.$$

Therefore, there is no potential and electric field inside the sphere. As for the potential outside, we know there will be an image charge of $-(a/d)q$ located at a^2/d for the conducting sphere. The potential from this image charge, expanded in spherical harmonics, is

$$-\frac{q}{4\pi\epsilon_0} \frac{a}{d} \sum_{l=0}^{\infty} \frac{r_{<}^l}{r_{>}^{l+1}} P_l(\cos \theta) = -\frac{q}{4\pi\epsilon_0} \frac{a}{d} \sum_{l=0}^{\infty} \frac{(a^2/d)^l}{r^{l+1}} P_l(\cos \theta) = -\frac{q}{4\pi\epsilon_0} \sum_{l=0}^{\infty} \frac{a^{2l+1}/d^{l+1}}{r^{l+1}} P_l(\cos \theta),$$

which agrees with the limiting form of A_l .